

MATH 5010 - Introduction to Mathematics of Finance, Fall 2025

*Exploring contemporaneous Trading Strategies centred on Term
Structure in Cryptocurrency Markets*

Instructor:
Dr Smirnov

Members (25Fall, MATH5010):

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Rehearsal:

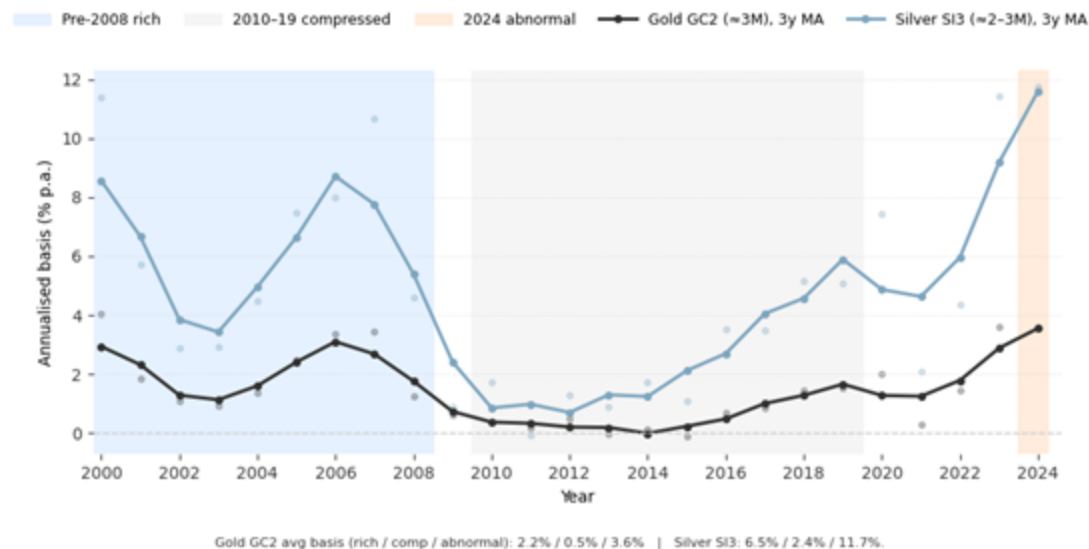
Dec-05-2025 10am (Zoom), All members present with Pietro

Tools Used: Python, Excel

Statement of Problem

Term-structure trades in traditional futures are widely seen as crowded and fee-compressed. Meanwhile, Cryptocurrencies like Bitcoin have become a derivatives-first market with large and persistent curve dislocations. This project tests whether two classic strategies, or “workhorses”, the basis trade and calendar-slope mean reversion, still deliver net-of-cost edge in BTC, clearly benchmarked against gold and silver, under a consistent framework for “apples to apples” comparison.

Compressed vs rich carry regimes in precious metals (context benchmark)



Research Question

Can traditional term-structure strategies still “work” in BTC, relative to gold and silver, once roll rules and transaction costs are applied?

Approach & Objective

Build two “vanilla” strategies that target the term structure:

- 1) Spot-Future Basis
- 2) Calendar Basis mean Reversion

Objective & Contribution

We study whether straightforward term structure strategies in crypto, such as basis trades, calendar spreads and funding carry, can generate economically meaningful risk-adjusted returns once realistic costs and implementation frictions are included, with explicit IS/OOS to avoid overfitting.

Methodology & Design (End to End)

1. Literature Review (Rates & Commodities)

Basis/carry, roll-down, calendar-spread mean reversion; why edges compress when trades get crowded. Industry Practitioners consulted. Initially evaluate, NatGas, WTI, Gold, Silver; Gold & Silver selected for similar economics.

2. Crypto Term-Structure Context

BTC is derivatives-first and funding-driven; curves can stay in contango and dislocate for long periods.

3. Research Question

Do classic curve trades still work in BTC, net of costs, versus traditional benchmarks (Gold, Silver)?

4. Data (Bloomberg)

XBT/USD spot + CME BTC futures; COMEX Gold (GC) and Silver (SI). Use the first 3 listed contracts, daily settlement prices.

5. Contract Mechanics & Cleaning

Compute DTE calendars based on rules, align panels, apply stitching rules, and enforce forced-close/roll thresholds.

6. Strategy Implementation & Iterate after initial presentation

- **S1 Spot–Futures Basis:** enter on high annualised basis; USD-notional matched long spot / short futures; exit on convergence or forced close.
- **S2 Calendar Mean Reversion:** trade front–next slope z-score; symmetric long/short calendar positions; exit on mean reversion or forced close.

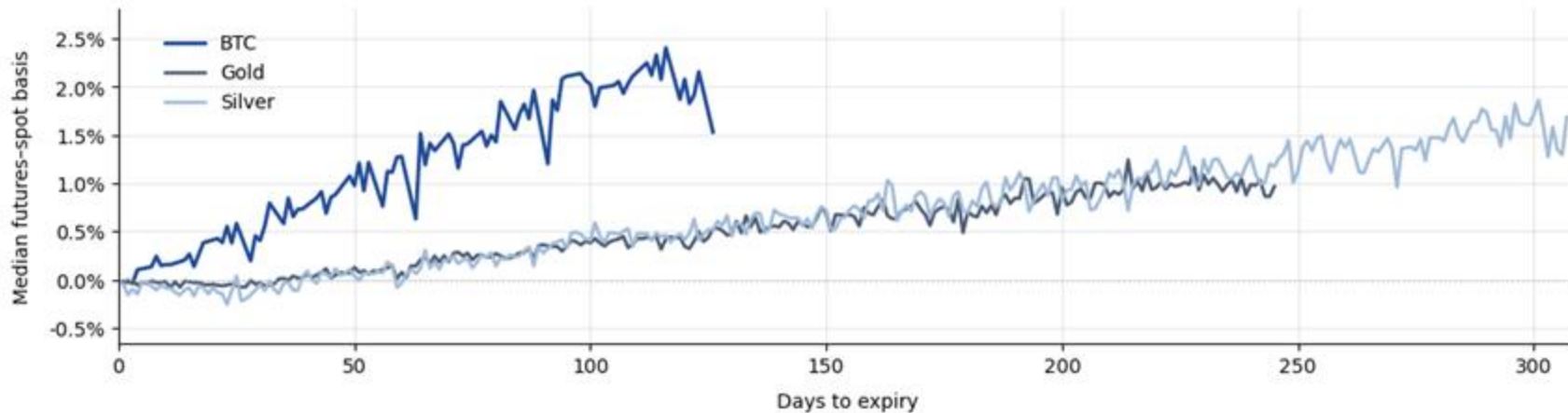
7. Backtest & Evaluation (IS/OOS)

In-sample 2018–2022, out-of-sample 2023–2025. Report annual return/vol, Sharpe, max drawdown, Calmar, and VaR/CVaR; compare BTC vs GC/SI.

Economic Intuition & Overview

Across BTC, gold and silver, term structures are usually in contango. Notably, BTC's front-end futures basis are structurally larger than precious metals, and is consistent with funding-driven demand, higher arbitrage frictions and lesser liquidity for leveraged long trades. Naturally, this manifests as more carry opportunity, but also greater drawdown risks.

Median % Futures–Spot Basis (Non-annualised) by DTE: BTC, Gold, Silver (Front 4



Note: First four listed futures only. BTC: monthly contracts (~30 days per tenor); Gold: bi-monthly (Feb/Apr/Jun/...) (~60 days per tenor); Silver: Mar/May/Jul/Sep/Dec (~70-80 days per tenor).

Precious Metals

- Gold and silver sit in **mild, stable contango** under a classic cost-of-carry framework.
- Basis is **small and smooth**, anchored by storage/financing costs and deep arbitrage liquidity.
- Mispricing is limited, making metals a **benchmark for traditional term-structure behavior** with lower-magnitude carry/roll opportunities.

Source: Bloomberg, Yahoo Finance

BTC

- BTC also trades in contango, but the **front-end basis is structurally richer** due to market structure, leverage demand, and convenience of futures.
- **Shorting frictions and funding constraints** slow arbitrage, allowing the basis to persist.
- Attractive for **short-maturity carry/roll**, but strongly **regime-dependent**, basis can compress or flip during deleveraging, such as during FTX collapse.

Strategy S1: 2-3M Basis Trade

We trade the 2-3 month spot-futures basis. When the annualised basis is sufficiently rich, we go long BTC spot and short BTC futures in USD-notional matched size, targeting basis compression with rule-based exits and forced rolls near expiry. We take the same notional long-short position, such that the trade is delta-neutral, and PnL is derived from basis decay

Signal Generation

$$basis_{k,t}^{ann} = \left(\frac{F_{k,t} - S_t}{S_t} \right) \frac{365}{DTE_{k,t}}, \quad k \in \{2M, 3M\}.$$

- **Entry/exit:** Enter when $basis_{k,t}^{ann} > \tau$, exit when $basis_{k,t}^{ann} < \tau$, or when $DTE_{k,t} < 3$ (7 for GC & SI)

Mark to Market Position

$$V_t = N_{spot} S_t - N_{fut} F_{t,T}$$

$$N_{spot} S_t \approx N_{fut} F_{k,t} \quad (USD - notional matched).$$

- MTM can swing even if convergence holds, so interim drawdowns and margin stress are possible.

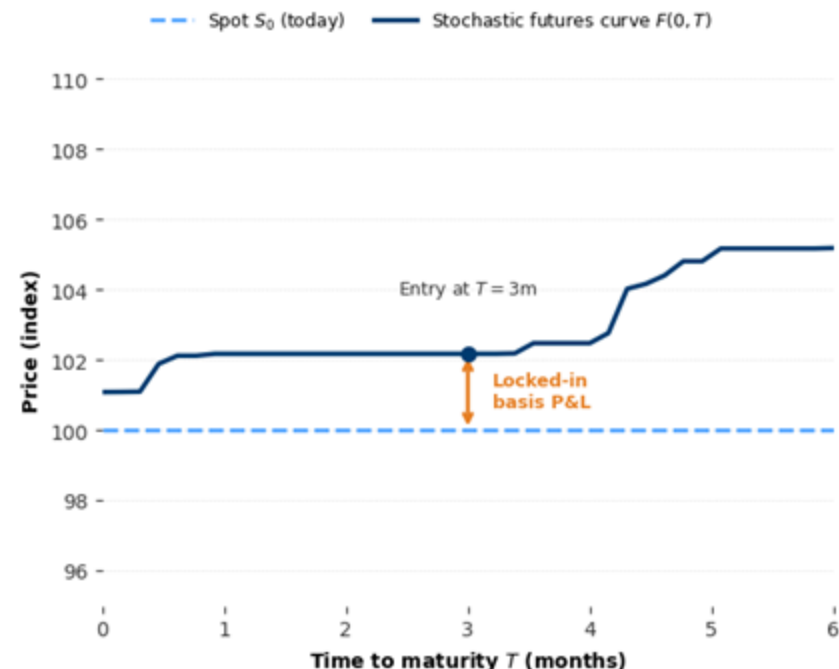
Incremental PnL

$$\Delta \Pi_t = N_{spot} \Delta S_t - N_{fut} \Delta F_{t,T} - TC_t$$

$$F_{k,T} \approx S_T \Rightarrow \Pi_T \approx N_{fut} (F_{k,0} - S_0) - \sum_{t=0}^{T-1} TC_t.$$

- P&L comes from **basis tightening**: futures falls relative to spot as expiry approaches.
- We report results under **multiple cost assumptions** (bps-style vs tick-style) since execution differs by venue and stress.

Sample Basis Decay



Strategy S1: 2-3M Basis Trade (BTC) Low-Risk Carry from Futures Term Structure

Core BTC term-structure sleeve harvesting 2–3M futures basis with limited directional risk.

Commentary

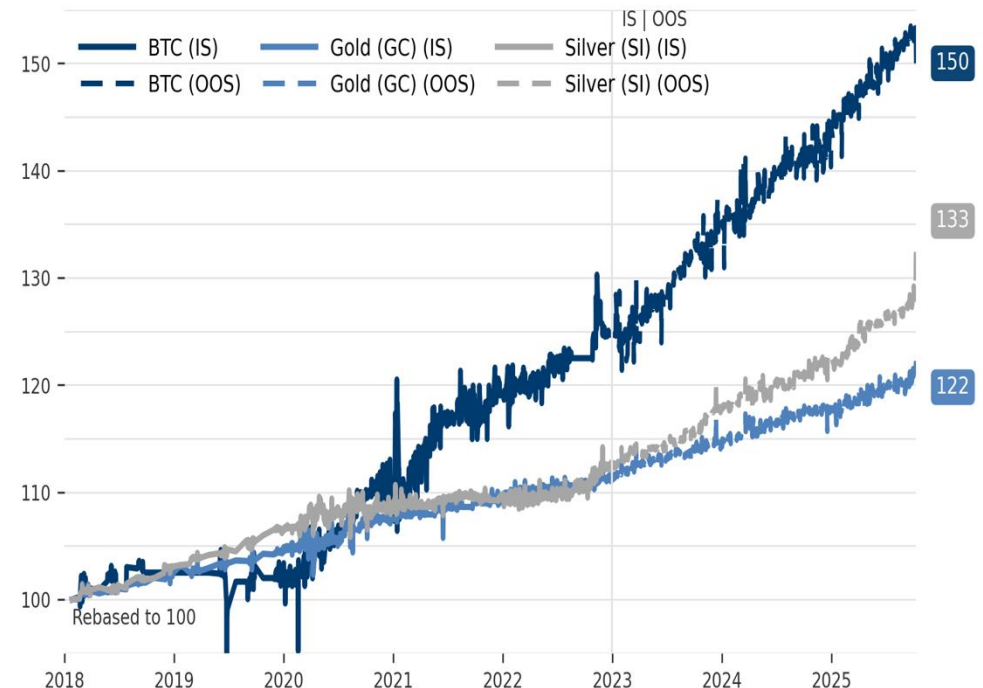
- Enter when annualised basis b_t^{ann} exceeds a threshold (BTC: 0.05, GC/SI: 0.02), exit when basis falls below τ_{exit} or on forced roll. Constructed as USD-notional matched long spot / short 2–3M futures.
- Performance from BTC suffered during the late 2019-20 from regime changes, funds piling on.
- In sharp deleveraging, like during FTX collapse, contango reverts to backwardation, the rule-based filter prevents trade.

Performance Metrics

	BTC Basis	Gc Basis	SI Basis
Annualised Return	6.41%	3.12%	4.45%
Volatility	19.57%	6.51%	6.57%
Sharpe Ratio	0.42	0.51	0.70
Maximum Drawdown	-17.6%	-4.7%	-3.94%
Sortino Ratio	0.67	0.83	1.18
Calmar Ratio	0.36	0.66	1.33

Source: Bloomberg

BTC vs Gold vs Silver - Roll-locked Carry (2nd Month)



BTC trade did not actually execute during FTX crash, when there was “negative basis”, i.e. $S > F$

Strategy S2: Calendar Spread Mean-Reversion (Near vs Next)

This Term-Structure Strategy seeks to monetise persistent contango in BTC futures by systematically harvesting mid-curve richness, while limiting drawdowns through disciplined entry/exit rules based on z-scored basis signals across rich, normal and poor curve regimes. GC & SI will be computed for a side-by-side comparison.

Calendar Slope

$$Cal_t = \frac{F_{next,t} - F_{near,t}}{F_{near,t}}.$$

Signal Construction:

- Standardize, with rolling window $L = 126$ to capture regime shifts (crypto + metals).

$$z_t = \frac{Cal_t - \mu_L}{\sigma_L}, \quad L = 126.$$

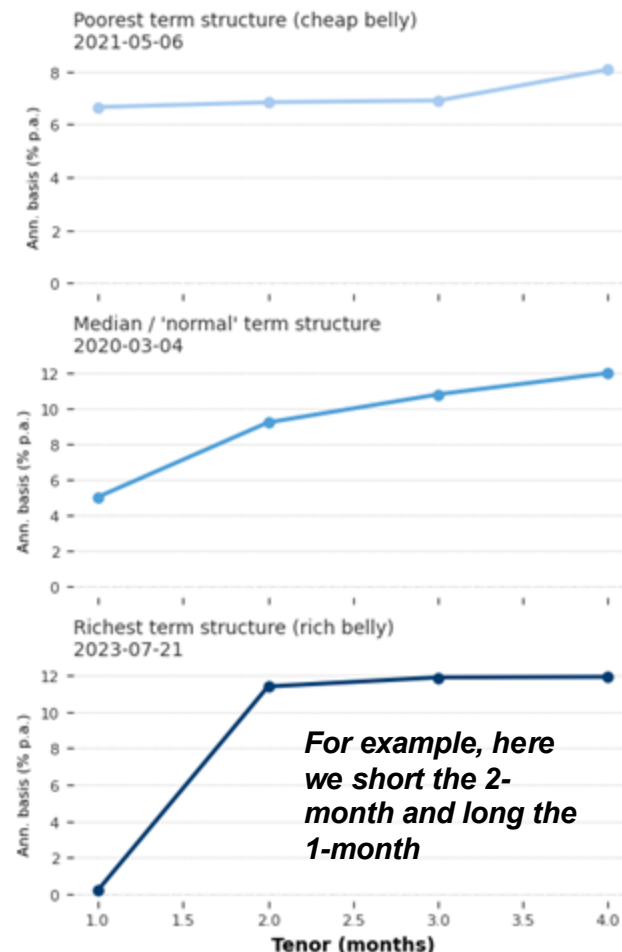
Trading Logic:

- Enter** when $Z \geq 1.6$ on any spread and **exit** when $Z \leq 0.2$ or when the near tenor rolls.
- Trade futures vs futures only:** long near / short far for rich curves (and vice versa);

$$\Delta \Pi_t = N_{near} \Delta F_{near,t} + N_{next} \Delta F_{next,t} - TC_t,$$

$$N_{near} F_{near,t} \approx N_{next} F_{next,t}.$$

Sample Term Structure



Strategy S2: Calendar Spread Z-Score (Near vs Next)

Calendar-slope mean reversion works best in metals; BTC works but is more regime-dependent.

Commentary

- Silver leads on risk-adjusted performance, with the strongest Sharpe/Calmar, indicating the cleanest convergence capture in this setup.
- BTC delivers comparable, but with noticeably higher volatility and deeper drawdowns, reflecting more variable term-structure regimes, especially post 2020.
- Forced rolls are handled mechanically (BTC: DTE < 3; GC/SI: DTE < 7), so results reflect tradable implementation rather than holding through expiry

Performance Metrics

	BTC Basis	Gc Basis	SI Basis
Annualised Return	2.246%	0.77%	2.403%
Volatility	2.609%	0.558%	1.57%
Sharpe Ratio	0.864	1.377	1.521
Maximum Drawdown	-2.857%	-1.01%	-1.172%
Calmar Ratio	0.786	0.762	2.05
VaR 95% Daily	-0.24	-0.023	-0.043

Source: Bloomberg

S2 Equity Curve (Near-Next Calendar Spread)

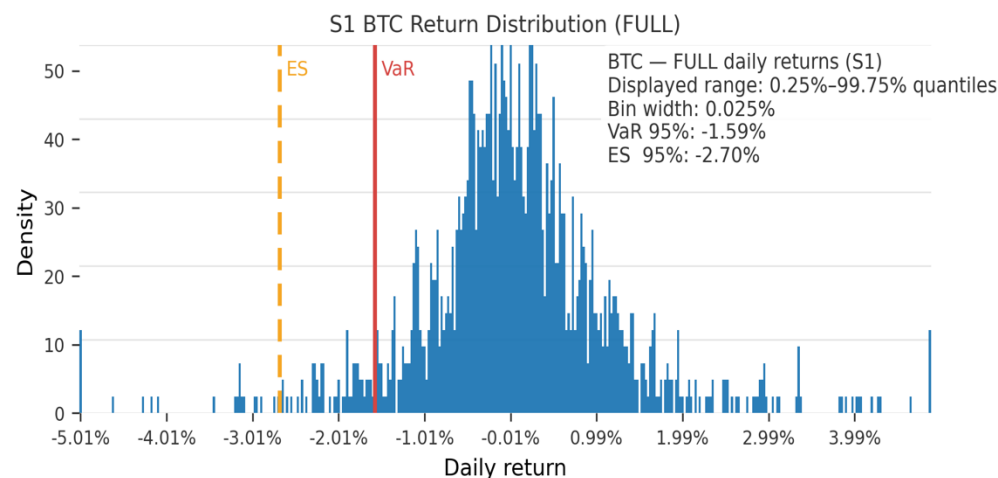
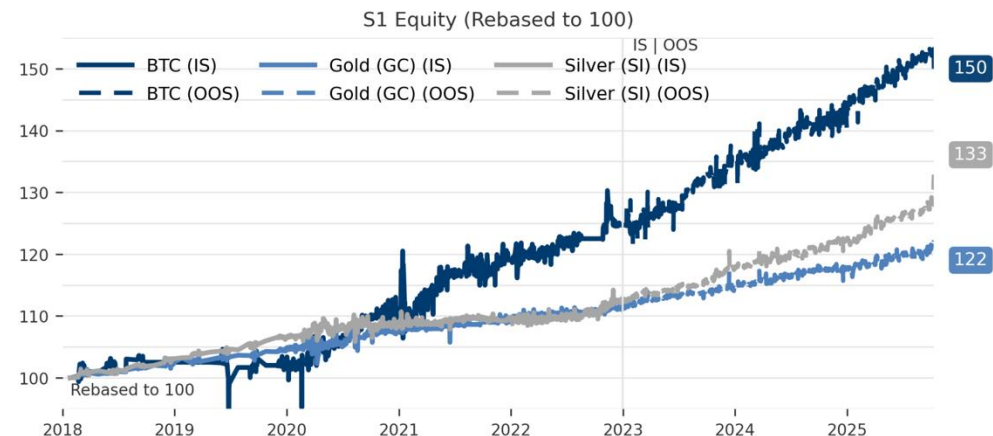


Evaluations

BTC offers structurally richer front-end carry than GC/SI, but realised outcomes are dominated by regime shifts, drawdowns, and implementation frictions

Evaluation: What we learn?

- For S1, BTC compounds the fastest, but the line is “choppier” with visible shock drawdowns of up to 20% during the interim before maturity. GC/SI rise more smoothly, consistent with lower path risk.
- Both S1 & S2, and associated benchmarks continue to function OOS, but the gap between the best performer of BTC in S1 and benchmarks is largely explained by larger “carry”.
- For S1 returns distribution, daily returns are tightly centered, but the VaR/ES markers show that negative days dominate the tail risk, despite decent average returns. Notably it does appear that the returns distribution is relatively symmetric.
- Key Takeaway from Study:** S1 demonstrates that these once-reliable curve trades can still “live on” in BTC versus GC/SI, yet the edge is less stable and more regime-dependent, requiring robust margins and risk controls to survive interim drawdowns. However, with appropriate margin-control, drawdown implications of margin calls can be mitigated.



Conclusion & Future Refinements

Conclusion

- Workhouse curve trades still *“live on”* in BTC< especially in S1 (Basis Trade), whereas BTC’s front-end carry is structurally richer than GC/SI, and therefore more opportunity to capitalize on basis decays
- The BTC edge is *not “free alpha”* as the realized performance is path-dependent, with pre-maturity widening of the basis driving drawdowns, and margin stress, despite being nominally delta-neutral.
- S2 (Calendar-Mean Reversion) has a cleaner risk profile, with better *risk-adjusted returns*, but substantively lower absolute returns, with GC/SI behaving more like a stable benchmark with far smaller tails.
- Compared to precious metals, BTC *offers more term-structure opportunity*, but drawdowns is what limited the realised Sharpe.

Further Steps

- *Volatility Targeting*, where exposures are scaled dynamically based on realized volatility, where exposures are tapered when realized volatility increases, so BTC path risk and subsequent drawdowns can be minimized. This has however, transaction cost implications as slippage is naturally much higher for BTC compared to other commodities.
- *OU-Bridge based signal*, where convergence is conditioned on expiry, as a “probability of tightening of filter” to avoid entering when convergence is unlikely or dispersion is higher.
- Operational de-risking. BTC, like other Cryptocurrencies, have *unique asset risks* in the form of theft from custodians etc., which presents itself as a Credit Risk. As such, if such trade were to be implemented, spot can be expressed via ETFs (e.g., IBIT) where appropriate to reduce custody and key-management risk while preserving basis exposure in the long-position .